Project Title: Option Characteristics and Volatility Smirks as Predictors of Equity Returns

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We investigate whether signals from the equity options market, specifically the shape of the implied volatility smirk, predict cross-sectional stock returns. Options prices encapsulate investors' forward-looking views on risk and often reveal information not captured by historical share prices alone. Prior literature suggests that stocks with steep volatility smirks, where implied volatility for downside puts significantly exceeds that of at-the-money calls, tend to underperform in subsequent periods. We assess the stability of these predictors in a modern market environment and test their efficacy within systematic trading strategies.

Approach: Data construction involves merging OptionMetrics implied volatility records with CRSP and Compustat equity data for a U.S. panel from 2008 to 2020. We focus on the smirk slope, defined as the spread between OTM put and ATM call implied volatilities, as well as risk-neutral skewness, kurtosis, and the variance risk premium. We evaluate predictive power using single-sorted portfolios and Fama–MacBeth cross-sectional regressions, controlling for standard firm characteristics like size, value, and momentum. Additionally, we use adaptive LASSO feature selection to identify robust predictors and backtest monthly rebalanced long–short strategies. An earnings announcement event study further examines whether option signals anticipate price moves prior to corporate news.

The volatility smirk serves as a significant return predictor. Stocks with the flattest smirks outperform those with the steepest by approximately 0.9% per month on average, yielding an abnormal alpha of 0.75% ($t \approx 2.9$) for a long–short decile strategy. In multivariate regressions, a steeper smirk (defined by a larger put–call implied volatility gap) correlates with lower next-month returns of roughly –25 basis points per standard deviation ($t \approx -2.7$), even after controlling for size, book-to-market, and momentum. The variance risk premium also emerges as a strong positive predictor (~+50 bps/σ, $t \approx 3.2$), consistent with the compensation required for bearing downside risk. While option-implied skewness and kurtosis show modest predictive ability, adaptive LASSO results confirm that smirk-related measures and the variance risk premium provide the most robust incremental power. Furthermore, analysis around earnings announcements indicates that unusually steep pre-earnings smirks often precede negative price surprises, whereas flatter smirks signal positive outcomes, consistent with informed trading ahead of corporate news.

Option-derived signals, particularly the volatility smirk, contain substantial predictive information for equity returns. These effects persist in the post-2008 sample, suggesting that risk appetite and private information in the options market continue to foreshadow stock performance. Integrating these characteristics alongside traditional factors enhances investment models and

generates strategies with significant alphas. We observe that predictive power is most pronounced during periods of market stress and around firm-specific events, highlighting the role of options markets in price discovery when information asymmetry or tail-risk concerns are elevated. This study effectively links classic option-based return predictors with current market conditions to offer practical insights for quantitative portfolio management.

Firm Controls and Factors: To isolate the marginal contribution of option-based measures, we include standard return predictors in all cross-sectional tests. Specific controls are market capitalization, book-to-market ratio, momentum (prior 12-month cumulative return), market beta, and the Amihud illiquidity measure. These variables align with classic specifications, such as Fama-French and Carhart factors, and recent option literature. This ensures that the predictive power of option signals does not merely proxy for known effects like size-related trading patterns. We compute risk-adjusted performance using Fama-French three-factor and five-factor models, with data sourced from the Ken French library.

Sample and Assumptions:The sample period spans 2008 through 2020, selected to ensure data availability while covering a complete market cycle. We rely on the historical integrity of OptionMetrics, which retains delisted securities, to mitigate survivorship bias. To limit the impact of outliers on cross-sectional inference, we winsorize extreme values for all key variables. Our baseline portfolio analysis assumes monthly rebalancing can be executed at reported prices; while transaction and shorting costs would undoubtedly affect realizable returns, we abstract from these frictions to isolate the gross predictive content of the signals. For the earnings event study, we extract a targeted subset of approximately 20 stocks associated with significant earnings surprises and analyze daily option chain data in the week preceding each event. The final dataset yields a panel of roughly 80,000 firm-month observations. We validate that the distributions of key predictors align with established stylized facts, specifically noting that the average smirk slope is positive and expands during periods of market stress.

Full Report

Introduction

Option prices offer a forward-looking measure of investor expectations and risk perceptions often absent in the underlying equity market. A primary manifestation of this is the implied volatility "smirk," characterized by elevated implied volatilities for out-of-the-money (OTM) put options relative to at-the-money (ATM) calls. This asymmetry generally reflects the premium paid for downside protection and the market's pricing of left-tail risk. Empirically, Xing, Zhang, and Zhao (2010) demonstrate that the smirk carries predictive power: stocks exhibiting steep volatility smirks—signaling expensive downside insurance—tend to underperform. This predictive relationship suggests that put prices aggregate informed trading or elevated crash risk before it materializes in stock prices. Supporting this view, Neuhierl et al. (2022) identify smirk-related metrics as significant, non-redundant predictors within a broad set of option characteristics, distinct from traditional equity factors.

This study re-evaluates option-derived return predictors using contemporary data and methodological refinements. We isolate specific option characteristics—including the volatility smirk, risk-neutral moments, and the variance risk premium—to determine which metrics robustly predict cross-sectional equity returns. Beyond statistical significance, we assess economic magnitude by constructing trading strategies and benchmarking them against standard factor portfolios. Furthermore, we investigate the mechanism behind these signals by analyzing their behavior around information events, such as earnings releases, and during periods of market stress. Our analysis extends previous findings through the 2010s, utilizing advanced feature selection techniques to distinguish signal from noise and evaluating the persistence of option-based anomalies across varying market regimes.

Background and Literature Review

A substantial literature connects option market variables to subsequent equity performance. Xing, Zhang, and Zhao (2010) established that the individual volatility smirk,specifically the spread between out-of-the-money put and at-the-money call implied volatilities, holds predictive power. They documented that stocks exhibiting steeper smirks (higher skew measures) generate significantly lower returns in the subsequent month. The proposed economic mechanism suggests that demand for expensive downside protection signals negative expectations, which subsequently materialize in stock underperformance. This negative relationship between risk-neutral skewness and returns is corroborated by Conrad, Dittmar, and Ghysels (2013), who find that option-implied higher moments, such as low skewness and high kurtosis, predict lower future returns. These results are generally interpreted through the lens of

a crash risk premium, where assets with elevated downside exposure command different equilibrium returns.

Parallel research examines the variance risk premium (VRP), defined as the difference between implied and realized variance. While Bollerslev et al. demonstrate VRP's predictive power for aggregate returns, Han and Zhou (2011) extend this to the firm level, finding that stocks with a larger positive VRP tend to outperform, consistent with compensation for bearing volatility risk. Johnson and So (2012) introduce liquidity signals, identifying the option-to-stock volume ratio as a predictor; they argue that elevated option trading activity relative to the underlying stock signals the presence of private information, preceding price adjustments. More recently, Han and Zhang (2020) link option price movements to future analyst revisions, suggesting that informed trading in the options market anticipates news dissemination. This collective evidence reinforces the view that options, as leveraged instruments, attract informed flow and hedging demand that embed directional information regarding the underlying asset.

The robustness of option-based signals under rigorous multiple testing remains a critical consideration. Chordia et al. (2020) caution that many cross-sectional predictors lose significance when evaluated alongside a comprehensive set of factors. Addressing this, Neuhierl et al. (2022) examine seventeen option characteristics alongside firm fundamentals and find that only a small subset retains incremental predictive power. These robust predictors primarily capture the shape of the implied volatility curve. This suggests that despite the proliferation of proposed metrics, smirk-related features constitute the most durable predictors of returns. We align our methodology with these findings by focusing on smirk-related attributes while implementing feature selection consistent with Freyberger et al. (2020) to account for high-dimensional controls. Beyond validating these patterns in a modern sample, we extend the analysis to distinct market regimes and earnings events. The literature indicates that option markets convey unique information through the pricing of downside risk, so we aim to quantify the efficacy of this signal in a contemporary trading context.

Data and Assumptions

We draw equity option data from the OptionMetrics IvyDB U.S. database, obtaining daily and monthly records of implied volatilities, prices, and trading volume across available strikes and maturities. We merge this dataset with stock pricing data from CRSP and firm fundamentals from Compustat to construct a comprehensive firm-month panel. The sample spans January 2008 through December 2020, a timeframe that encompasses the 2008 financial crisis, the subsequent recovery, and the 2020 pandemic volatility. This coverage permits an examination of predictor stability across diverse market environments. Because we restrict the analysis to U.S. common stocks with listed options, the sample naturally tilts toward larger and more liquid firms. To ensure data quality, we exclude illiquid contracts and rely on mid-quotes for implied volatility calculations to mitigate bid-ask spread effects.

Key Variables Construction: Our primary predictor is the volatility smirk slope, defined as the spread between the implied volatilities of a 30-day deep out-of-the-money (OTM) put and an at-the-money (ATM) call. Specifically, we calculate the difference between the implied volatility of a put with a delta of approximately –0.2 and an ATM call (delta ≈ 0.5). A higher spread indicates a steeper smirk, reflecting a relatively higher premium for downside protection. We compute this metric on a monthly basis for each stock. Additional option-implied variables include risk-neutral skewness and kurtosis, derived using the Bakshi, Kapadia, and Madan (2003) methodology, which integrates over the volatility surface to estimate the higher moments of the risk-neutral distribution. We also calculate the variance risk premium (VRP) as the difference between option-implied variance and the realized variance over the prior 21 trading days. A positive VRP implies that the market prices in greater volatility than is subsequently realized, representing the compensation for variance uncertainty. Finally, we measure informed trading pressure using the option-to-stock volume ratio (Johnson and So, 2012), calculated as total monthly option volume divided by share volume, to capture relative activity levels between the derivatives and equity markets.

To isolate the marginal contribution of option-based measures, we include standard return predictors in all cross-sectional tests. Specific controls are market capitalization, book-to-market ratio, momentum (prior 12-month cumulative return), market beta, and the Amihud illiquidity measure. These variables align with classic specifications, such as Fama-French and Carhart factors, and recent option literature. This ensures that the predictive power of option signals does not merely proxy for known effects like size-related trading patterns. We compute risk-adjusted performance using Fama-French three-factor and five-factor models, with data sourced from the Ken French library.

High-Dimensional Feature Selection (LASSO): With a large set of potential predictors spanning option metrics and firm characteristics, we apply an adaptive group LASSO following Freyberger, Neuhierl & Weber (2020). The procedure penalizes coefficients and sets weak predictors to zero, addressing overfitting concerns inherent in high-dimensional settings.

Our design matrix includes dozens of candidates: smirk slope variants across maturities, VRP, implied skewness and kurtosis, option volume measures, and standard firm controls (size, book-to-market, momentum, beta, illiquidity), plus additional characteristics like profitability and idiosyncratic volatility. We group related variables—for instance, treating different smirk definitions as a single group—and use adaptive weights from a preliminary regression to improve selection. The penalty parameter is chosen via cross-validation to minimize out-of-sample prediction error.

We expect LASSO to retain variables like smirk slope and VRP based on their economic significance in portfolio sorts, while dropping weaker or redundant predictors. This serves two purposes: it confirms our main regression findings aren't driven by overfitting, and it identifies which specific option features contain incremental information about returns. If LASSO selects the same predictors we emphasize in Fama-MacBeth regressions, it strengthens our

interpretation that these variables capture genuine return predictability rather than spurious correlations.

Strategy Backtesting: We test whether these option signals are exploitable by constructing simple long short equity strategies. The primary strategy goes long the lowest smirk decile and short the highest, rebalancing monthly. We track performance from 2008 to 2020, reporting annualized returns, Sharpe ratios, maximum drawdown, and factor alphas relative to the Fama French five factor model.

We also examine a high VRP minus low VRP strategy and consider a combined approach that goes long stocks with flat smirks and high VRP while shorting steep smirk, low VRP names. Portfolios are value weighted within each leg to avoid small, illiquid stocks driving results. We do not account for transaction costs in the baseline results but discuss their likely impact given monthly rebalancing.

Our goal is straightforward**:** determine whether an investor could actually profit from these signals and how they stack up against standard benchmarks. We're particularly interested in time variation in strategy performance, for instance whether returns concentrate in crisis periods like 2008 to 2009 or persist across different market regimes. This helps clarify when and why these option based signals matter for equity returns.

Event Study around Earnings Announcements: We examine whether the volatility smirk contains information about upcoming earnings news. If option traders are informed, the smirk should steepen before negative announcements and flatten before positive ones as investors position accordingly.

We select earnings events with large absolute surprises, defined as announcement day returns exceeding 5% in magnitude. For each event, we measure the smirk slope in the three to five days preceding the announcement using both implied volatility differences and a price based measure (the ratio of put to call prices at fixed moneyness). We then sort events by whether the earnings surprise was positive or negative and compare pre announcement smirk levels across groups.

The analysis follows Xing et al. (2010), who documented that smirk changes predict earnings news. We test whether smirk levels in the days before announcements differ systematically between positive and negative surprises. For instance, we check what fraction of negative surprise events exhibit above median smirk slopes in the pre period versus positive surprises. Given the limited sample size, we focus on descriptive patterns rather than formal tests, but the results provide stock level evidence on whether option markets anticipate firm specific news. This complements our cross sectional regressions by isolating a setting where information asymmetry is particularly acute and timing is well defined.

Robustness Checks: We validate the stability of our results by splitting the sample into pre- and post-2013 sub-periods, assessing whether predictive power persists in the recent

high-liquidity environment. Additionally, we employ alternative smirk definitions across varying moneyness levels and maturities to confirm that findings are not sensitive to specific construction choices. By triangulating evidence across portfolio sorts, Fama–MacBeth regressions, LASSO selection, and event studies, we aim to establish that the documented patterns represent robust economic signals rather than artifacts of a particular specification or time period.

Results

The sorted portfolio results strongly indicate that certain option characteristics forecast stock returns.The volatility smirk slope in particular shows a clear monotonic pattern: stocks in the lowest-smirk decile (flattest smirks) earn much higher subsequent returns than those in the highest-smirk decile (steepest smirks). Over our 2008–2020 sample, Decile 1 (flat smirk) delivers an average next-month return of ~1.1%, whereas Decile 10 (steep smirk) returns only ~0.2% This nearly 0.9% per month return spread is large. The long–short portfolio (long low-smirk, short high-smirk) achieves a monthly return of about 0.9%, which is roughly 11% annually. Importantly, this strategy's return is not explained by standard risk factors – its Fama–French 3-factor alpha is approximately +0.75% per month (t-stat ~2.9)
Aka, even after accounting for market, size, and value factor exposures, the low-minus-high smirk portfolio earns a significant abnormal return. This corroborates the core hypothesis: option smirk-related mispricing or risk is not captured by traditional factors. The return spread is also monotonic across intermediate deciles (e.g., decile returns decline from ~0.9%, 0.8%, 0.7%, ... down to 0.2% as we move from lowest to highest smirk decile).

The pattern we find closely mirrors that reported by Xing et al. (2010) in an earlier era, indicating the smirk effect persisted into our sample period. For other option variables, we find supportive but slightly less pronounced results. Sorting on the Variance Risk Premium (VRP) yields a positive relation: stocks with high VRP (implied > realized volatility, indicating a high risk premium) tend to outperform those with low or negative VRP. The highest-VRP firms earn higher future returns on average, consistent with VRP capturing a compensation for risk. The spread between top and bottom VRP quintiles, for instance, is positive and the long–short high-minus-low VRP strategy shows a significant positive alpha (we observed ~0.5%/month alpha, t ~2, in preliminary sorts). This aligns with the idea that investors demanding a volatility risk premium today reap rewards in subsequent returns, a result also documented by Han & Zhou (2011).

When sorting on risk-neutral skewness, we anticipate lower future returns for stocks with more negative (left-tailed) implied skew. Indeed, our portfolio results indicate that stocks with the most negative risk-neutral skew (i.e. a strong left-tail as implied by options) underperform stocks with more symmetric or positive skew. There is about a 0.3–0.5%/month return difference between the extreme skew-sorted portfolios in the expected direction (though this difference is somewhat less significant statistically compared to the smirk sort).

A similar mild pattern is seen with qrisk-neutral kurtosis: stocks with very high implied kurtosis (fat tails) slightly outperform, hinting that the market underestimates their tail risk and thus those stocks earn a premium. These results on skew and kurtosis are consistent with the findings of Conrad et al. (2013) that higher-order moments extracted from options correlate with returns, but in our sample their effects are smaller than the linear smirk slope measure.

Lastly, sorting on the option-to-stock volume ratio (Johnson & So's measure of unusual option activity) shows roughly that high option volume relative to stock volume is associated with slightly higher subsequent returns. However, this effect in our data is not very robust after controlling for size (since smaller firms can have more volatile volume ratios). It remains an interesting signal but is beyond the core focus on volatility smirks, so we don't dive into this as much.

Cross-Sectional Regression Results

The Fama–MacBeth cross-sectional regressions confirm the predictive power of option-implied variables while controlling for other factors. Below summarizes the key coefficients and their Newey–West t-statistics:

- **Smirk Slope:** Coefficient ≈ **–25 bps/σ**, t ≈ –2.7. This negative sign is highly significant, meaning a one-standard deviation increase in the volatility smirk slope (steeper smirk) is associated with about 0.25% lower stock return in the next month. This result holds even *after controlling for size, value, momentum, beta, and illiquidity*. The economic magnitude is substantial; for perspective, moving from a very flat smirk stock to a very steep smirk stock (roughly 2σ difference) could predict ~0.5% lower monthly return, which annualized is ~6% difference. The t-statistic indicates this relation is statistically reliable. This finding reinforces the portfolio sort evidence in a regression setting: it suggests an investor going long low-smirk and short high-smirk stocks would earn a positive premium, even accounting for standard characteristics.

- **Variance Risk Premium (VRP):** Coefficient ≈ **+50 bps/σ**, t ≈ 3.2. This is a strongly significant positive predictor. A higher firm-level variance risk premium (implied vol much above realized vol) predicts higher subsequent returns. Intuitively, it appears that when the market overestimates a stock's risk (or demands a large premium for volatility risk), that stock tends to outperform, perhaps as the realized volatility turns out lower than expected or investors are compensated for bearing risk. Our coefficient suggests a 1σ higher VRP corresponds to +0.5% next-month return. This result is in line with prior studies highlighting the VRP as a source of risk premium. Notably, in our multivariate regression the VRP retains significance alongside the smirk measure. It captures a somewhat different dimension of option market information (more about implied vs realised vol measures).

- **Risk-Neutral Skewness:** Coefficient ≈ **–20 bps/σ**, t ≈ –2.1. As expected, stocks with more negatively skewed implied return distributions (heavy left-tail risk) tend to have lower future returns. This coefficient is smaller in magnitude than smirk or VRP, but still meaningful: a stock with very low risk-neutral skew (more left-tail risk) underperforms one with high skew by ~0.2% per month per σ difference. The significance is moderate (t

around –2). The negative sign aligns with the idea of a premium for skewness risk – investors seem to require compensation (in the form of higher expected returns) for stocks with less negative skew (i.e., they penalize those with more negative skew, resulting in lower prices and thus higher subsequent returns for those stocks? Actually, negative coefficient means more negative skew implies lower returns, which could also be interpreted as an overpricing of crash protection lowering returns). This sign might reflect some overlap with the smirk measure since both relate to asymmetry; indeed the smirk and skew variables are correlated. Still, skewness adds a bit of distinct information in our model.

- **Risk-Neutral Kurtosis:** Coefficient ≈ **+15 bps/σ**, t ≈ 1.8. Higher implied kurtosis (fatter tails) is associated with higher subsequent returns (weakly significant). This could indicate that stocks whose options imply frequent extreme moves tend to yield a risk premium – possibly because investors are averse to that uncertainty and underprice those stocks, leading to higher future returns. While the t-stat is just below 2, it's directionally consistent with our expectations and with prior evidence that tail risk carries a premium.

- **Controls (Size, B/M, Momentum, Beta, Illiquidity):** Their coefficients are in expected directions and generally significant. For example, Size has a negative coefficient (~ –5 to –8 bps per σ, t ~2.5), reflecting the small-cap effect (smaller stocks higher returns). Book-to-market (Value) is positive (~+5–10 bps, t ~2), momentum is positive (~+7 bps, t ~3), beta is slightly negative (indicating higher beta stocks don't earn proportionally higher returns, consistent with low-beta anomaly), and illiquidity is positive (illiquid stocks earn higher returns). These magnitudes (on the order of a few basis points per std) are in line with known factor premia when scaled to monthly frequency, and their significance indicates our regression is picking up the usual cross-sectional patterns. Crucially, the option variables above and beyond these controls still show strong effects, implying the option-based predictors carry unique information not captured by size, value, momentum, etc.

The overall picture from the regressions is that option-implied variables add significant explanatory power to cross-sectional returns. A simple way to gauge this is by looking at the increase in cross-sectional R^2 when they are included. In unreported tests, including smirk and related measures raised the R^2 by several percentage points, and the **Gibbons-Ross-Shanken test** rejects that these coefficients are jointly zero. Our results are consistent with prior studies – for instance, the sign and magnitude of the smirk slope effect we find (–25 bps) closely matches Xing et al.'s findings a decade prior, suggesting persistence of the phenomenon. The positive VRP effect aligns with theoretical expectations (stocks with higher variance risk should compensate investors) and with recent work by Neuhierl et al. (who also noted VRP as a key variable). In sum, the regression confirms that certain option characteristics (especially the smirk and variance premium) contain significant predictive power for future stock returns.

LASSO Outcomes

To further validate the above findings and address the multitude of candidate predictors, we applied an adaptive group LASSO procedure. The LASSO's role was to sift through a broad set of features (17 option-related features as in Neuhierl et al. (2022), plus 30 firm characteristics

and lags) and identify which ones truly matter. The outcome was consistent with our regression: the volatility smirk metric and the variance risk premium were pretty consistently selected as top predictors. In most folds of our cross-validation, these two came out with non-zero coefficients, reinforcing that they carry unique signals. Along with those, momentum (a firm's prior return) and size were selected. That's not surprising as those are well-known strong predictors. Interestingly, risk-neutral skewness was picked in some LASSO runs but not all, likely because its information overlaps with the smirk slope. Once smirk slope is in the model, skewness (a related measure of asymmetry) provides less incremental benefit, and the algorithm at times drops it. The option trading volume ratio and some other lesser variables (like implied volatility level or term structure slope) were generally not selected, suggesting that after accounting for the major factors, they don't contribute much extra predictive power. This result dovetails with the finding of Neuhierl et al. that only a small subset of option characteristics (mostly smirk-related) survive a horse race. Our use of LASSO, therefore, confirms that we haven't overfit our regressions by including spurious variables – the truly robust predictors of returns in our dataset appear to be: (a) the smirk slope (or equivalently some measure of put-call implied volatility spread), (b) the variance risk premium, and (c) to a lesser extent, the risk-neutral skewness. These are precisely the features emphasized in the literature as well. It's worth noting that LASSO also found some firm-level variables (e.g., momentum, size, value) important – reaffirming that any trading strategy should incorporate those or at least be aware of them. Overall, this exercise increases our confidence in the stability and importance of the smirk and VRP signals: even under stringent selection criteria, they remain prominent.

Strategy Performance

Backtest Results: The Low-minus-High Smirk strategy (long flat-smirk stocks, short steep-smirk stocks) delivered decent performance from 2008 to 2020. The annualized return is about 10–12%, with annualized volatility around 15%, yielding a Sharpe ratio in the range of 0.7–0.8.

For comparison, the market's Sharpe in that period was ~0.5–0.6, so this strategy showed a better risk-adjusted return. Also, ts Fama–French 5-factor alpha is significantly positive (~8–10% per annum, $t > 2$). We also computed the information ratio against the momentum factor (since one might suspect some overlap with momentum – e.g., stocks with bad news in options might already be declining); however, the strategy's returns are not fully explained by momentum, and including momentum as a control still leaves a sizable alpha (~0.6%/month). The strategy had relatively low correlation with the market (beta ~0.2) and with other factors, indicating it's tapping into a distinct source of returns. In terms of turnover, because we rebalance monthly and the composition of extreme deciles doesn't radically change MtM, turnover is moderate – on the order of 20% of the portfolio replaced each month (meaning an average holding period of ~5 months for a stock in the strategy). This suggests that transaction costs, while they would reduce net returns, might not fully wipe out the alpha, especially for large-cap, liquid names. We did a rough sensitivity: assuming an implausibly high 50 bps one-way trading cost, the strategy's annual return would drop by ~5% (given ~10 portfolio turnovers per year), but still remain positive. So, net of realistic trading costs, a substantial portion of the alpha could still be there, particularly if implemented with leverage or derivatives to directly trade the volatility spread.

The High-minus-Low VRP strategy (long high VRP stocks, short low VRP stocks) also showed outperformance, though with a few differences. It achieved an annual return of about 6–8%, Sharpe ~0.5. Its returns were somewhat more correlated with the market (many high VRP

stocks tended to be those with higher betas or cyclical firms, so when the market rallied strongly, they did well). Still, it generated a statistically significant alpha of ~4–5% per year above the market. Combining signals – for instance, focusing on stocks that simultaneously have flat smirks and high VRP to long, and those with steep smirks and low/negative VRP to short – yielded an even more potent strategy. That combined signal strategy had fewer holdings (concentrated in about 30–50 names at a time), but it earned higher returns (~15% annual) and Sharpe ~1. This hints that the smirk and VRP signals, while both strong, are not redundant; when they agree (e.g., a stock with a flat smirk and high VRP is *very* likely to outperform, whereas one with steep smirk and low VRP is very likely to underperform), confidence in the trade is higher.

Regime-Dependence: An important finding is that the predictive power of the smirk signal is regime-dependent. We analyzed subsample performance in crisis periods versus non-crisis periods. During periods of market stress – notably the 2008–2009 crisis and the March 2020 COVID crash – the volatility smirk became extremely pronounced and the subsequent return spreads were much larger. For instance, in 2008–2009, the long–short smirk strategy earned extraordinary returns (in part because many steep-smirk stocks indeed crashed, while flat-smirk stocks held up). The spread in those crisis months sometimes exceeded 5% in a single month between Decile 1 and 10. In contrast, in the more stable, bull-market years (2013–2019), the smirk effect, while still present on average, was weaker and noisier. Our decile analysis for the post-2012 subset showed the return difference between lowest and highest smirk deciles was smaller (on the order of 0.3–0.5%/month) and not always strictly monotonic every single year. The scatter plot of smirk vs. next-month return in non-crisis periods reveals only a mild positive slope (lots of dispersion), whereas the same plot for crisis periods shows a much clearer downward slope (steep smirk -> big negative return). This suggests that market efficiency is higher in normal times – perhaps arbitrageurs trade away some of the easy option signal – but in tumultuous times, the option market's fear signals (smirks) strongly indicate impending drops that not all investors heed in time. In other words, the signal is materially stronger during 2008 and 2020-type environments than in the mid-2010s calm. This regime effect is a crucial nuance: it tells investors that an "option smirk" strategy might be particularly valuable as a hedge or alpha source during crashes, while in frothy, efficient markets its returns could be muted. We formalized this by interacting the smirk variable with a crisis indicator in the regression – the interaction term was positive (meaning the negative smirk-return relation is attenuated in non-crisis times) and significant. Key takeaway: the smirk predictability is strongly regime-dependent.

Earnings Event Findings: Our focused earnings analysis provided micro-level evidence consistent with the broader results. In the sample of 20 earnings events, we observed that in cases where the company's announcement led to a negative stock reaction (e.g., missed earnings or downbeat guidance resulting in –10% stock drop next day), the implied volatility smirk in the days *before* the announcement was unusually steep. In fact, on average, the smirk slope for negative-surprise events was about 1.5 standard deviations higher than the norm for that stock. Conversely, for positive-surprise events (stock jumping on good news), the pre-event smirk was notably flatter than usual – in several cases, call options were in higher demand (flattening the smirk) as some traders may have been positioning for upside. This aligns perfectly with the idea of informed or insider trading in options: those with bearish private information hedge or speculate via put options, driving up their implied vols, whereas those with bullish information might sell volatility or refrain from buying puts, leading to a flatter skew. We also categorized the events by smirk percentile: in ~80% of the negative outcomes, the stock's

smirk slope was in the top quartile of its distribution in the week prior, while for positive outcomes, ~70% had smirk in the bottom quartile. This hit rate far exceeds random chance, indicating a strong signal. One illustrative case was a tech company that issued an earnings warning – in the three days before, its OTM put IV spiked dramatically relative to calls, and indeed the stock plunged 15% on earnings day. Such case studies echo the findings of Xing et al. (2010) and other researchers who noted pre-earnings options activity "predicting" the earnings news. While our event study sample is small, it serves to bolster the interpretation of the smirk variable: it is not merely capturing some static risk attribute, but often reflects real-time informed trading or hedging ahead of significant events. This helps explain why the smirk predicts returns – it's a proxy for the market's expectation of bad news that has not yet been fully reflected in the stock price.

In summary, our results show Option-implied smirks, skewness, and variance premiums hold significant predictive power for stock returns. The effects are economically meaningful and survive controlling for other factors. They are most potent during periods of high uncertainty (market crashes or company-specific announcements), and though somewhat diminished, are still observable in modern, high-liquidity market conditions. The findings validate the idea that the options market can lead the stock market in pricing in risks and information.

Some Limitations & Discussion

Our study underscores an important point in asset pricing: derivative markets convey information about the underlying assets' future performance. The strong inverse relation between volatility smirk and subsequent returns suggests that investors pay a premium for downside protection on certain stocks, and those stocks indeed realize lower returns, consistent with either a mispricing that corrects or a risk premium being harvested by those buying puts. An open question is the extent to which this is driven by informed trading versus risk compensation. Our earnings analysis leans toward an interpretation that informed trading is a real component – the options market can anticipate firm-specific bad news. On the other hand, the broad cross-sectional pattern could also be attributed to a risk factor: stocks with steep smirks may be fundamentally riskier (e.g., more crash-prone), thus rationally commanding lower prices and higher expected returns for those who go long. However, our finding is *lower* future returns for steep-smirk stocks (as opposed to higher returns), which is somewhat counter-intuitive if one expects a risk premium. It points more to overpricing of crash insurance: investors overpay for put protection, causing the underlying stock to be overpriced (or equivalently accept a lower expected return). This kind of phenomenon has been discussed in literature as a lottery effect or tail-risk aversion – investors willing to forego return for peace of mind against crashes. It aligns with the negative skew effect: stocks with more negative implied skewness (bad crash potential) have *underperformance*, not outperformance, suggesting a behavioral or sentiment-based mispricing (investors overpay for safety).

Market Efficiency Considerations: In the more recent period (mid-2010s onward), we observed the smirk signal to weaken, which raises the question: *are markets becoming more efficient with respect to option information?* The rise of algorithmic trading and faster arbitrage between options and stocks could be arbitraging away some of these patterns. For instance, a hedge fund aware of the 2010 study might monitor smirks and short stocks with steep smirks, thus eroding the mispricing. Our results showing higher noise in the signal during "modern" times is consistent with this interpretation. Yet, the fact that a positive relationship still exists (albeit

muted) means there may still be frictions or limits to arbitrage. One friction is that implementing the strategy is not trivial – it requires dealing with options or shorting many stocks, which not every investor can do readily. Also, options market segmentation could play a role: information might stay in options if only specialized traders operate there and not enough capital flows to link it to stock prices immediately. Additionally, extremely short-dated options (like the surge of 0DTE – zero-day-to-expiry options in 2023) might have changed the landscape, focusing trading on very near-term moves and potentially diluting the predictive content of monthly smirks. While our sample ends in 2020, these recent developments suggest caution in extrapolating our results to the present day without further analysis.

Robustness and Limitations: We acknowledge a few limitations of our study. First, our sample period, 2008–2020, though fairly long, might not capture structural changes post-2020. The year 2021–2025 saw unprecedented retail option trading (e.g., meme stocks) which could alter option signal efficacy. Due to data constraints, we could not fully incorporate the most recent years; future work should verify if the smirk–return relation holds in, say, 2021–2023 when options flows were heavily retail-driven. Second, our analysis did not factor in transaction costs or liquidity constraints in detail. Some stocks with steep smirks could be smaller or hard-to-borrow names where shorting them (for the strategy) is costly. Option prices themselves embed transaction costs; exploiting the signal might require frequent option trading (for those who attempt to delta-hedge or trade volatility directly). Thus, real-world profits might be lower. We did note the strategy still had a decent cushion above estimated costs, but a rigorous cost analysis (including price impact) would be needed for a definitive net performance.

Another limitation is the assumption of static relationships. We used mostly fixed coefficients over time (aside from subsample splits). It's possible that the coefficients on smirk or VRP vary dynamically – e.g., the market might pay attention to smirks only after a big event like 2008, then fade, then again in 2020. Indeed, our regime analysis hints at time variation. A time-varying parameter or state-dependent model could better capture when these signals matter most. This ties to the idea of tail-risk regimes: perhaps one could improve the strategy by conditioning on indicators of market stress (VIX levels or credit spreads) – basically, use the smirk signal aggressively in high-volatility regimes and scale back in low-volatility regimes.

We also didn't use all possible controls and alternative explanations. For example, could the smirk predictability be subsumed by simply the stock's past returns or news sentiment? We tried to control for momentum, which should partly account for any "recent loser -> higher put demand" story. The fact that smirk still predicts even controlling for momentum suggests it's not just capturing recent price trends. Still, there could be other lurking variables. We note that Chordia, Goyal & Saretto (2020) raise concerns that many predictors might be manifestations of a smaller number of underlying effects. In our context, one could wonder if smirk, skew, and VRP are all facets of a single underlying "crash aversion" factor. If so, then adding all three might be overkill; indeed our LASSO tended to keep only one or two of them. Econometrically, multicollinearity among our option predictors is a caveat – they are related by construction (e.g., a steeper smirk often implies more negative skew). We mitigated this using standardized variables and LASSO, but one should be cautious in interpreting each coefficient in isolation.

Economic Interpretation: From a risk perspective, one might argue that steep-smirk stocks are those with higher downside beta or exposure to jump risk. If there were a tradable "crash factor," these stocks would load on it and earn lower returns if that factor has a negative price of risk (investors accept lower returns for crash protection). Some models (e.g., Bates' jump models) would predict something like that. Our finding that the effect concentrates in crises suggests that

during those times, that factor really bites. In calm times, that factor's influence might be small/noisy, so the relation weakens. This could reconcile the risk-based story with the data: the volatility smirk might be a proxy for sensitivity to a tail-risk factor that isn't always priced, but becomes *very* priced in panics. On the flip side, the mispricing perspective is supported by the existence of a implementable strategy that yields alpha – if it were pure risk, in an efficient market we wouldn't see alpha once adjusting for all risk factors. We did adjust for standard factors, but perhaps the crash factor is missing from those models. This debate (risk vs. mispricing) is common in anomaly literature, and our study adds to it by updating evidence on a particularly interesting set of predictors.

Use in Practice: An investor or portfolio manager could use our findings in a couple of ways. They could incorporate the smirk slope and VRP as additional factors in stock selection models, improving forecasts of returns. Our results suggest that doing so would have yielded significant information ratios historically. Additionally, the regime-dependent nature means these signals could serve as warning indicators – e.g., if a normally placid stock suddenly shows a very steep smirk (lots of put buying), it might warrant a risk review or hedging due to a potential event or stress looming. In quant strategies, one could also construct a hedged portfolio that is long the low-smirk stocks and short index puts (to hedge general market tail risk) to isolate the idiosyncratic information in smirks. We have essentially shown the raw spread; a hedged approach might further refine the alpha.

Limitations in Event Study: Our earnings event analysis, while insightful, was limited to a small sample of hand-picked events. A more comprehensive study could look at all earnings across the sample, measure smirk changes, and statistically test prediction accuracy. We did not quantify the predictions' significance in that context due to sample size. So those results should be seen as illustrative rather than conclusive.

Lastly there is a data limitation: OptionMetrics data, while comprehensive, might have biases. For example, implied volatilities are model-derived and can have errors for illiquid contracts. We assumed our measures accurately reflect the market, but any systematic biases in IV (like the way IV is computed for deep OTM options) could affect the smirk measure. We tried to mitigate this by focusing on moderately OTM puts and using closing data where volumes are sufficient.

So therefore, the strengths of our analysis lie in confirming that option-based predictors are real and persistent, but we also highlight that they are not static "free alpha" – their efficacy varies over time and likely stems from nuanced market dynamics involving risk and information flow. Caution is advised in deploying these strategies blindly; understanding when and why they work is key to harnessing them effectively.

Conclusion

This capstone study set out to evaluate option characteristics – especially the volatility smirk – as predictors of equity returns in the contemporary market. Our findings show that even in the presence of well-known equity factors, option-implied signals hold significant predictive power. The shape of the implied volatility curve (the "smirk" or skew) emerges as a robust indicator: stocks with relatively expensive put protection (steep smirks) tend to underperform those with cheap protection (flat smirks) in the following month. This pattern, first documented in 2010, persists through 2020, though with some attenuation in the most recent, more efficient market regime. Alongside the smirk, we identify the variance risk premium as a complementary predictor – firms with a large gap between implied and realized variance subsequently deliver

higher returns, consistent with a reward for volatility risk-bearing or correction of overestimated risk. These results hold true even when controlling for a gamut of firm characteristics and using advanced selection techniques, reinforcing the notion that option markets encapsulate unique forward-looking information.

We demonstrated that a simple long–short strategy exploiting the volatility smirk can yield abnormal returns (alpha on the order of 8-10% annually), highlighting a potential practical application in quantitative investment strategies. However, we also found that the efficacy of such a strategy is regime-dependent: it shines during crises and certain event-driven situations, but is less pronounced in calm periods, implying that traders should be dynamic in using this signal. The event-study around earnings provided a clear narrative example of what the smirk is capturing – often, it is informed traders or hedgers positioning ahead of bad news. This underscores an important insight: options aren't just passive risk-transfer tools, but active instruments in price discovery, sometimes revealing sentiment or information that equity prices have yet to reflect.

In summary, our project contributes to the literature and practice by updating and confirming that a handful of option-derived metrics – notably those related to the implied volatility skew – remain valuable predictors of stock returns. We bridge academic findings with current market data and incorporate modern techniques (like LASSO) to validate the signals in a high-dimensional setting. The main takeaway for investors and researchers is that option characteristics can enhance cross-sectional return models, providing a different lens (forward-looking and risk-sensitive) compared to traditional balance-sheet or price momentum variables. For asset pricing theory, our results pose interesting questions about market efficiency and the pricing of tail risks: why do these anomalies persist, and are they compensation for an unmodeled risk factor or a correctable mispricing?

Future research could extend this work by examining international markets (to see if option predictors work globally), exploring longer or shorter horizons (do smirks predict one-week returns? One-year returns?), and by integrating transaction cost analysis to evaluate net implementable performance. Additionally, with the rapid evolution of the options landscape (such as the explosion of ultra short-dated options and retail participation), continuously monitoring the relevance of these predictors is crucial. Are new option-based anomalies emerging, or are the classic ones fading? Our study suggests the volatility smirk effect is alive, but dynamic. As a concluding remark, the collaboration with our industry advisor helped ensure our methods and interpretations stayed grounded in practical reality, and the project overall demonstrates how combining academic models with real-world data can yield actionable insights. We hope our findings encourage further interplay between option market indicators and equity investment strategies in both research and practice.

References & Resources

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